# Zero-Day Detection under HAI Plant-Version Shift

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## The problem

A zero-day detector must flag attacks it has never seen, trained on normal operation alone. The harder case is when the *plant itself* is replaced by a later, differently instrumented version: HAI 21.03 and 23.05 are two such versions of the same testbed, sharing 59 tags but not the instrumentation.

**How much does detection degrade from 21.03 to 23.05, and can an adaptive decision layer recover what is lost?**

Thresholds are the obvious suspect. A detector that alarms at a fixed rate on one plant version will alarm at a different rate on the next, and an operator cannot retune it by hand. If the representation still separates attacks from normal operation, a threshold that follows the stream should recover the loss. If it does not, no threshold can.

## Data and protocol

| partition | files | rows | attacks |
|---|---|---|---|
| 21.03 normal training | 3 | 921,603 | 0 |
| 21.03 test | 5 | 402,005 | 50 |
| 23.05 test | 2 | 284,400 | 52 |

The two versions share 59 public tags; 18 have zero variance throughout the normal 21.03 training data and are removed *using source statistics only*, leaving **41 modelling tags**. Their constant source encoding would otherwise dominate max-fused scores after the shift.

## Protocol

- **No attack labels** are used for training, model selection, or threshold calibration.
- Normalisation is fitted on chronological 21.03 training partitions; the last 15% supplies normal calibration scores.
- EVT is calibrated on **source-normal** scores to an empirical target of **10 false alarms/day**. Three seeds (11, 23, 37); no point adjustment.

## Three detectors

- **PCA** — reconstruction error on flattened windows.
- **Deep SVDD** — a bias-free 1-D CNN trunk into a latent hypersphere (13,904 parameters).
- **Convolutional VAE** — the same trunk plus Gaussian heads and a decoder (26,352 parameters).

Causal 60 s windows, scored at the window end, 10 s evaluation stride. Per-sensor scores are fused by **maximum**, so an alarm needs one strongly anomalous sensor. The two neural models share the trunk but not the total parameter count, and that gap is reported rather than hidden.

## Result 1: in-domain, at 10 alarms/day

| | event recall | eTaF1 | PR-AUC | Hit@3 |
|---|---|---|---|---|
| PCA | 0.740 | 0.533 | 0.502 | 0.229 |
| Deep SVDD | **0.773** | **0.543** | 0.489 | 0.171 |
| VAE | 0.560 | 0.473 | 0.462 | **0.257** |

At 10 false alarms per day on 21.03 all three work moderately — and none of them works well. A PR-AUC near 0.5 at a 1–2% attack rate is a weak ranker: there was little left to lose.

## Result 2: the representation collapses

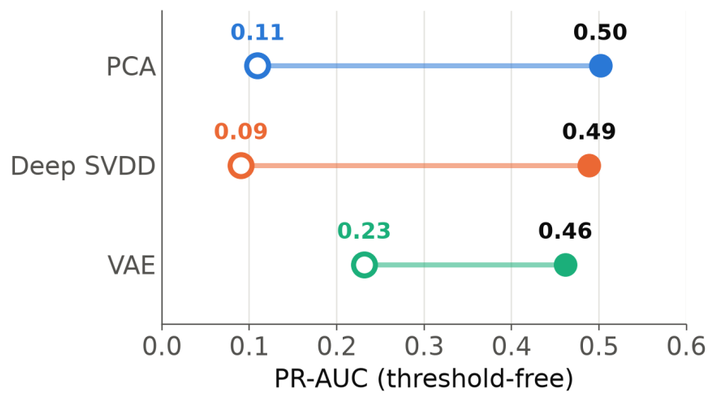



Figure 1: PR-AUC is threshold-free, so it measures the representation alone.

PCA 0.50 → 0.11, Deep SVDD 0.49 → 0.09, VAE 0.46 → 0.23. VAE retains the most; PCA and Deep SVDD retain almost nothing.

## Result 3: recall 1.000 is alarm collapse

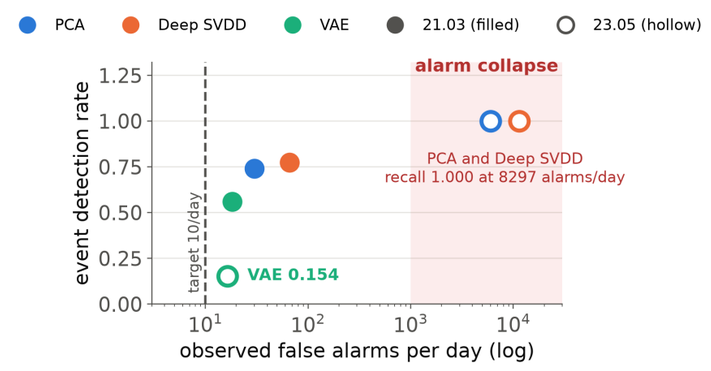



Figure 2: Frozen arm; the calibration target was 10 alarms/day.

On 23.05, PCA and Deep SVDD report **event recall 1.000** because they fire **8,297 alarms per day**, on 96% of normal windows. A circular-shift null settles it: a randomly time-shifted alarm series matches this recall in 200 of 200 shifts ($p = 1.00$), while on 21.03 the same models beat their null at $p < 0.001$.

## Result 4: DSPOT cannot recover it

| model | frozen | adaptive | gain | FA/day |
|---|---|---|---|---|
| | *23.05 event recall* | | | *frozen→adapt.* |
| PCA | 1.000† | 1.000† | 0.000 | 8297→8297 |
| Deep SVDD | 1.000† | 1.000† | 0.000 | 8297→8297 |
| VAE | 0.154 | 0.115 | **−0.038** | 16.4→5.5 |

†Alarm collapse: a circular-shift null reaches 1.000 in 200/200 shifts ($p = 1.00$). Not detection.

**Why.** PR-AUC is *identical* across arms: both threshold the same scores. DSPOT moves the threshold, not the encoder, the PCA basis, or the score.

## Conclusions

- **Decision-layer adaptation cannot substitute for representation quality.** DSPOT helps where the representation still works and does nothing where it does not.
- **Zero-day detection does not survive the version shift.** Ranking quality falls four- to five-fold; two of three detectors collapse into continuous alarm.
- **Report threshold-free metrics beside event recall**: recall 1.000 and 0.154 can describe equally useless detectors.
- Limits: one testbed, two versions, three seeds; the source calibration targets an empirical source rate, not a guaranteed shifted-domain budget.

## References

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[3] L. Ruff et al. *Rethinking Assumptions in Deep Anomaly Detection*. arXiv:2006.00339, 2020.
[4] A. Siffer et al. *Anomaly Detection in Streams with Extreme Value Theory*. KDD 2017.
[5] W.-S. Hwang et al. *Time-Series Aware Precision and Recall for Anomaly Detection*. CIKM 2019.
[6] W.-S. Hwang et al. *Do You Know Existing Accuracy Metrics Overrate Time-Series Anomaly Detections?* SAC 2022.
[7] S. Kim et al. *Towards a Rigorous Evaluation of Time-series Anomaly Detection*. AAAI 2022.